

Math 2320 - FE 1.4 Review

1. $\int \frac{1}{x^2 - 1} dx$
2. $\int \frac{2x - 3}{x^2(2x + 3)} dx$
3. $\int \frac{-5}{x(x^4 + 4)} dx$
4. $\int \frac{-5}{x(x^4 + 4)} dx$
5. $\int \frac{x^4}{x^2 - 1} dx$ Perform long division first

$$\begin{array}{r}
 x^2 + 1 \\
 x^2 - 1 \overline{) \quad x^4} \\
 \underline{-x^4 + x^2} \\
 x^2 \\
 \underline{-x^2 + 1} \\
 1
 \end{array}$$

So

$$\frac{x^4}{x^2 - 1} = x^2 + 1 + \frac{1}{x^2 - 1}$$

6. $\int \frac{x^3}{x^2 + 2x + 1} dx$ Perform long division first (and then factor the denominator prior to partial fractions)
7. $\int \frac{x^3}{(x^2 - 1)^3} dx$ Trig sub or partial fractions could work here. The denominator has a degree of six, so for Partial fractions. That is we need A, B, ..., F. Maybe Trig sub is easier?

8. Graph the following conic sections.

- (a) $x^2 + y^2/2 = 1$
- (b) $x^2 - y^2 = 1$
- (c) $x + y^2/2 = 1$
- (d) $x + y^2/2 = -1$
- (e) $x^2 + y^2/2 = -1$
- (f) $(x - 1)^2 + (y - 3)^2/4 = 4$
- (g) $(x + 1)^2 - (y - 1)^2/4 = 1$

9. Translate the following equations from rectangular to parametric.

- (a) $y = x^2 + x$
- (b) $x = y^2 + y$
- (c) $x^2 + y^2 = 4$
- (d) $x^2 + y^2/16 = 1$

10. Translate the following equations from parametric to rectangular.

- (a) $x = 3t - 1$ $y = 2t + 5$
- (b) $x = t^2 + 1$ $y = t$
- (c) $x = 2t - 1$ $y = t^2 + 1$
- (d) $x = \cos(t)$ $y = \sin(t)$
- (e) $x = 2 \cos(t)$ $y = 3 \sin(t)$
- (f) $x = t \cos(t)$ $y = t \sin(t)$ This may be ugly

11. Graph the following equations given in parametric form.

- (a) $x = 3t - 1$ $y = 2t + 5$
- (b) $x = t$ $y = t^2 + 5$
- (c) $x = \cos(t)$ $y = \sin(t)$
- (d) $x = 2 \cos(t)$ $y = 3 \sin(t)$
- (e) $x = t \cos(t)$ $y = t \sin(t)$